

Tuple = It is ordered , can contain duplicates values, and cannot be changed after creation.

- Tuple can hold mixed data types.
- Iterable
- Allows duplicates
- Immutable

```
In [1]: x = ("yasir", 20, 7.7)
        print(x)
```

```
('yasir', 20, 7.7)
```

```
In [2]: name, age, cgpa = x
```

```
print(x)
print(name)
print(age)
print(cgpa)
```

```
('yasir', 20, 7.7)
yasir
20
7.7
```

```
In [3]: print(x[0])
        print(x[-1])
```

```
yasir
7.7
```

```
In [4]: print(len(x))
```

```
3
```

```
In [5]: print(x.count(20))
        #count how many times the element is repeted
```

```
1
```

```
In [6]: print(x.index("yasir"))
        #finding the position of the elements
```

```
0
```

```
In [7]: z = ("yasir")
        print(z[0:3])
        print(z[-3:-1])
        print(z.find("yasir"))
        print(z*3)
```

```
yas
si
0
yasiryasiryasir
```

```
In [8]: a = ("yasir")
        b = ("arafat")
        print(a+" "+b)
```

yasir arafat

```
In [9]: # Convert a list to a tuple.
```

```
x = [1,2,3,4]
print(tuple(x))
```

(1, 2, 3, 4)

```
In [10]: # Convert a tuple to a list and modify an element.
```

```
x = (1,2,3,4,4)

print(list(x))
```

[1, 2, 3, 4, 4]

```
In [11]: print(x.count(4))
```

2

```
In [12]: print(x.index(2))
```

1

```
In [13]: t = (1,2,3)
        x,y,z = t

        print("x :", x)
        print("y :", y)
        print("z :", z)
```

x : 1
y : 2
z : 3

```
In [14]: # Create a tuple of numbers and find their sum.
```

```
t = (1,2,3)
print(sum(t))
```

6

```
In [15]: # Create a tuple of numbers and find their sum.
```

```
t = (1,2,3)
x = max(t)
y = min(t)
print(x)
print(y)
```

3

1

```
In [16]: t = (1,5,3)
        print(tuple(sorted(t)))
```

(1, 3, 5)

```
In [17]: # Merge two tuples and remove duplicates.
```

```
a = (1,2,3,4,4,5)
b = (1,5,5,6,7,4)

x = tuple(set(a + b))
print(x)
```

```
(1, 2, 3, 4, 5, 6, 7)
```

```
In [18]: # Check if all elements in a tuple are the same.
```

```
t = (1,1,1)

same = len(set(t)) == 1
print("all element is same : ", same)
```

```
all element is same : True
```

```
In [19]: # Create a nested tuple and access an element from the inner tuple
```

```
t = (1,2,(1,2,3),4,5)

x = t[2][1]
print(x)
```

```
2
```

```
In [ ]:
```